Alchemy system design v0.4.0 – Wizard Yan's Diary: The Remembrance Alchemy

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Case study summary

Design Challenge

Replace recipe-driven crafting with system where players learn elemental properties through experimentation and discovery.

Approach

Model ingredients as movement vectors, then build a minimal 1D prototype to test whether players can infer the underlying rules.

Preliminary result

Three exploratory playtests suggested that players could reason about elemental displacement, while revealing issues with the understanding of the interface.

Iteration

Introduced tutorial and interaction changes in prototype 0.3.0 in response to observed problems.

System overview

Purpose

Alchemy is the primary knowledge-based system, through which the player learns to understand and manipulate elemental properties. Rather than collecting recipes, the player experiments with materials, observes outcomes, and gradually learns the principles governing their interactions. When designing it, I wanted to deconstruct the usual crafting trope of “collect materials, choose recipe, click craft”.

Core Alchemical Principle

Alchemy is a spatial reasoning system. Ingredients are charged with elemental energy, which moves the mixture through elemental space. The player learns to construct a trajectory toward the desired outcome.

Design goals

The rules should be simple and learnable. The system should not punish the player for failed experiments or require them to start over in case of a mistake or misclick. The outcomes should be deterministic. Each problem should have multiple valid solutions. Complexity should gradually increase throughout the game. The player should be provided with all the knowledge they need for a solution. The system is primarily a puzzle game.

Player mental model

The player should think of alchemy as navigating a space. When they choose ingredients, they pick movement patterns. When processing ingredients, they modify those patterns. They might try constructing a path from start to destination or the other way around. Depending on the topology they might need to avoid certain areas, which forces them to try a different combination of ingredients. Assuming that the length of a path is bounded, the player will have to search for a more efficient solution. If a path is not constructible with currently unlocked ingredients, they might need to search for new ingredients in other systems.

Model

Elemental space – Alchemical map

The game uses the five classical elements: fire, water, air, earth and Aether. They are color-coded: red for fire, blue for water, yellow for air, brown for earth and violet for Aether. The elemental space is a 3D coordinate system, in which the axes represent elemental energy: water – fire on horizontal axis and air – earth on the vertical axis. Aether energy corresponds to depth axis. A useful way to think about the elemental space is a set of 2D layers (maps), like floors of a skyscraper. At the beginning of an alchemy experiment, all elements are in balance – the point (0,0,0). Adding ingredients moves the mixture in three dimensions, which can be thought of as navigating a labyrinth. The number of ingredients per mixture is fixed but can be expanded further in the game (see Fuel section in this document).

Discovery zones are convex 2D subsets of layers. For sake of simplicity, they can be thought of as rectangular shapes covering some vertices. Landing the mixture in a discovery zone unlocks the corresponding discovery. These can be new ingredients, alchemical stations or their upgrades (i.e. capacity).

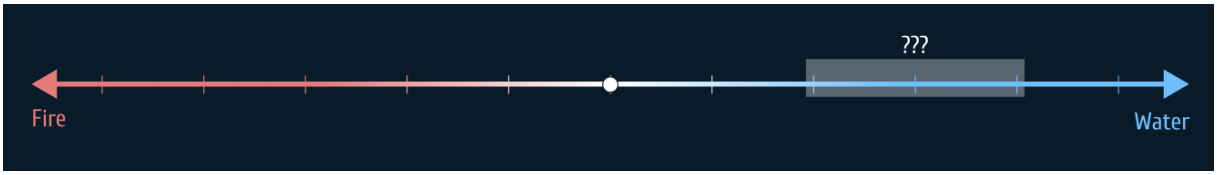
The game introduces elemental space first as 1D space (line), then adds more dimensions as the story progresses.

Water and Fire - First dimension

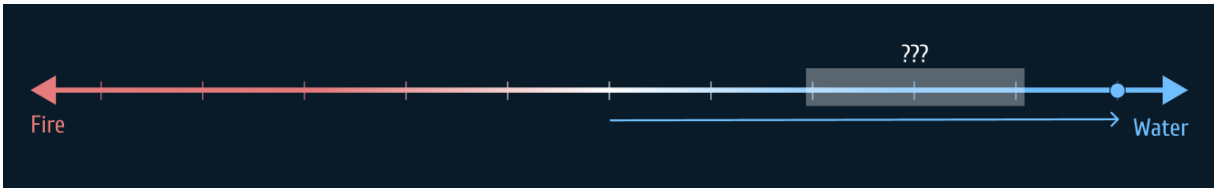
In the first dimension, the elemental space is a number line with water on the right end and fire on the left end. The experiment moves along a single axis, starting at 0. Ingredients push the mixture toward fire or water depending on their composition. 1 water energy corresponds to moving 1 unit right, 1 fire energy corresponds to moving 1 unit left. For an ingredient with composition (*water*, *fire*), the movement becomes (*water-fire*) units to the right. Discovery zones are intervals on number line, like [2,4]. When the mixture ends up with water energy between 2 and 4, the discovery is unlocked (see below for an example).

Assume the player has these ingredients at their disposal:

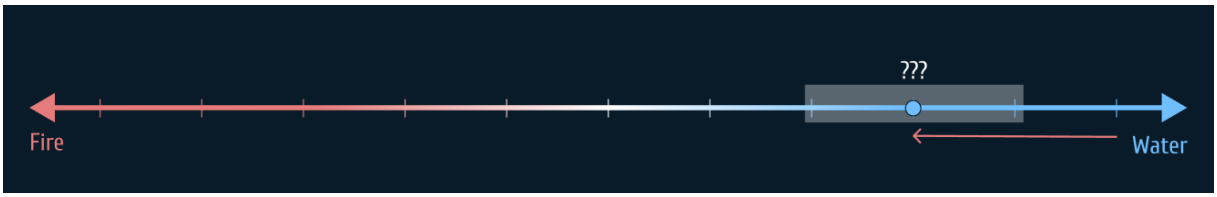
- **Water crystal** (5 water)
- **Steam** (2 fire)



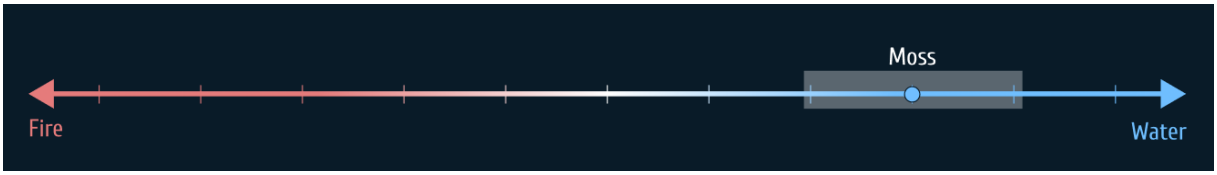
Initial state, mixture at 0.



The player adds a **Water crystal**, the mixture moves 5 units right (5 water).



The player adds **Steam**, which moves the mixture 2 units to the left.



The mixture lands in the discovery zone, **Moss** ingredient is unlocked!

Player model

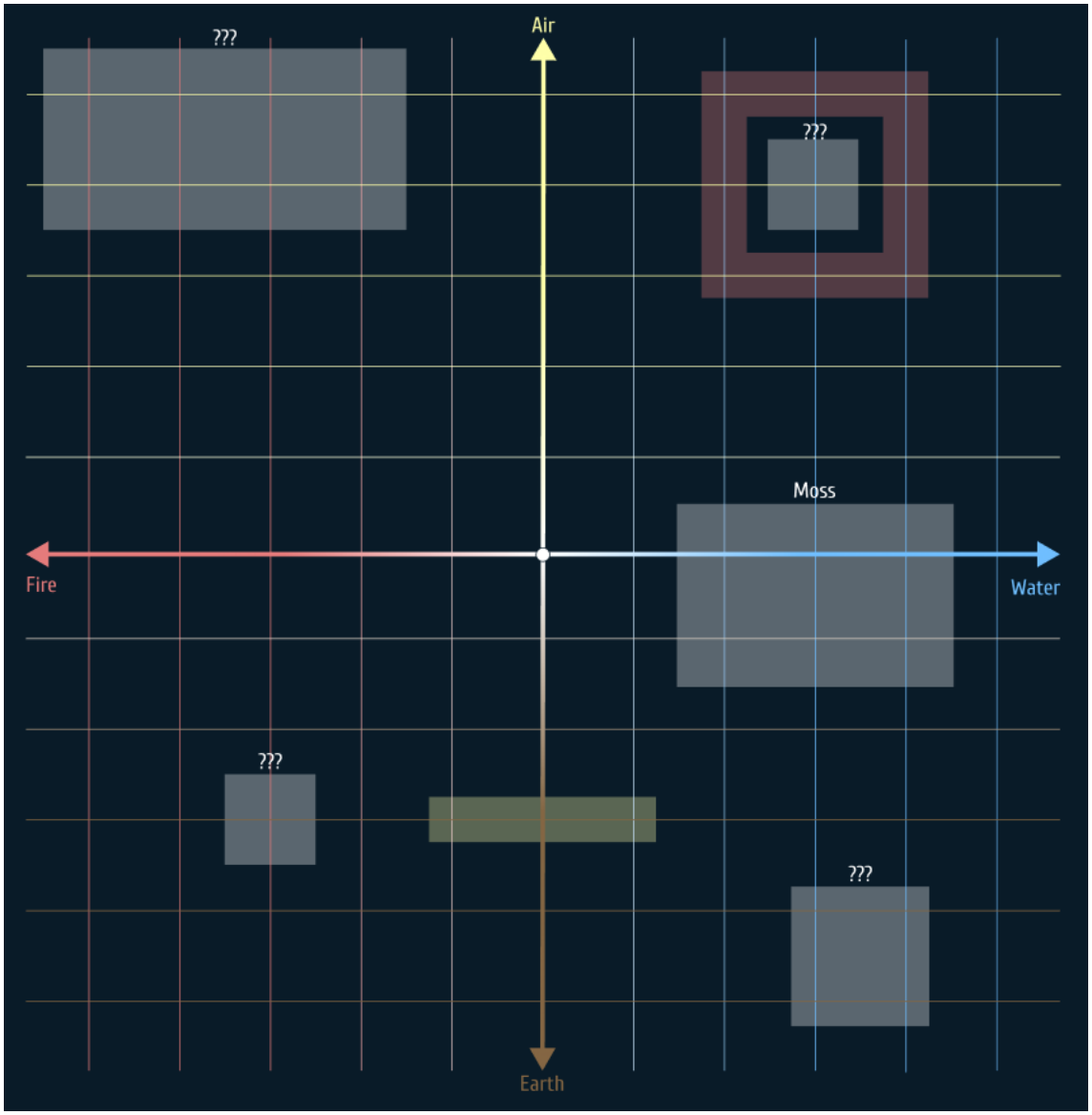
The player should notice the similarity between this one-dimensional elemental space and the number line. Both ends of this elemental space are color-coded for elements and evenly spaced notches indicate units. When the player sees an ingredient with composition (5 water, 0 fire), they should assume this moves the mixture 5 units to the right (the direction of water). Similarly, an ingredient with composition (0 water, 2 fire) should move the mixture 2 units to the left.

Then the player encounters an ingredient with both fire and water energy. They observe that the movement corresponds to a composition of two moves (to the left and to the right), effectively cancelling out each other.

Air and Earth axes - Second dimension

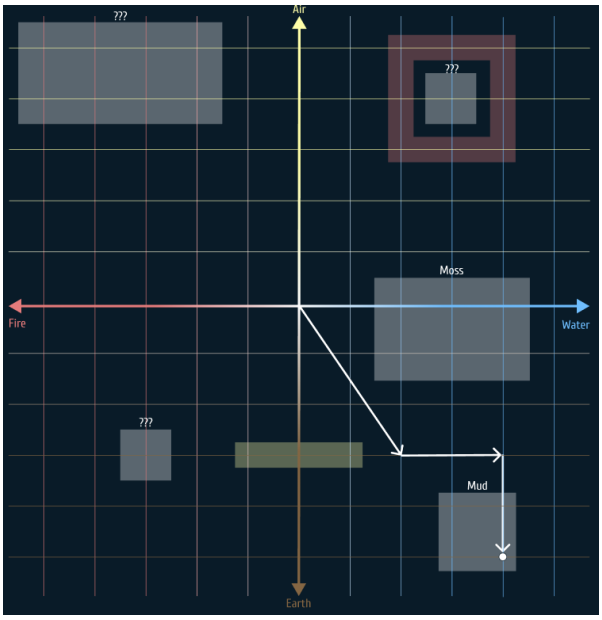
When the player learns the basic principles of movement and elemental cancelling, the game adds another dimension – Air/Earth axis. The elemental space becomes a 2D map of alchemy. Ingredients can now have four elemental energies. For instance, an ingredient with 2 water, 3 earth energy moves the mixture 2 units right and 3 units down.

The discovery zones become rectangles, which increases the solution space, as more coordinate pairs correspond to the same result. Later in the game, colorful zones will also appear on the map. Look at the example alchemy map below:

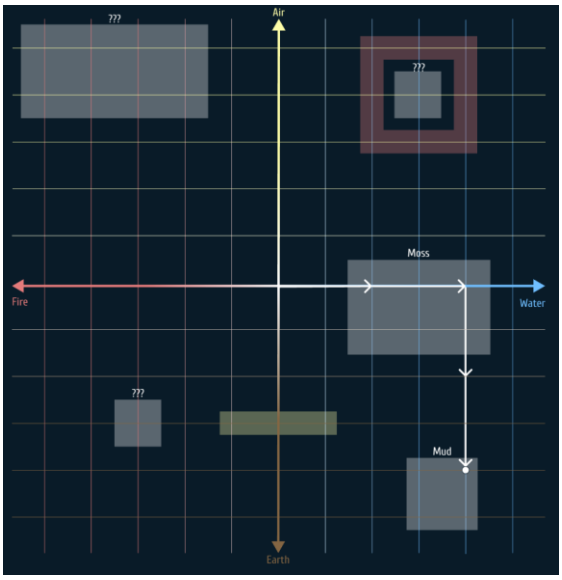


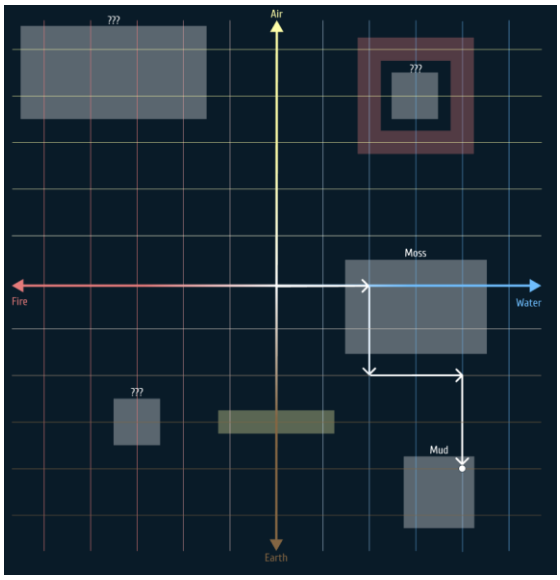
The experiment starts at the origin (0,0), where all elements are in balance. Player's goal is to arrive at any of the discovery zones using available ingredients. The ingredients can be thought of as vectors in 2D space. An ingredient with composition (*air, earth, water, fire*) becomes a vector (*water – fire, air - earth*).

They might construct a path like this to discover **Mud**:



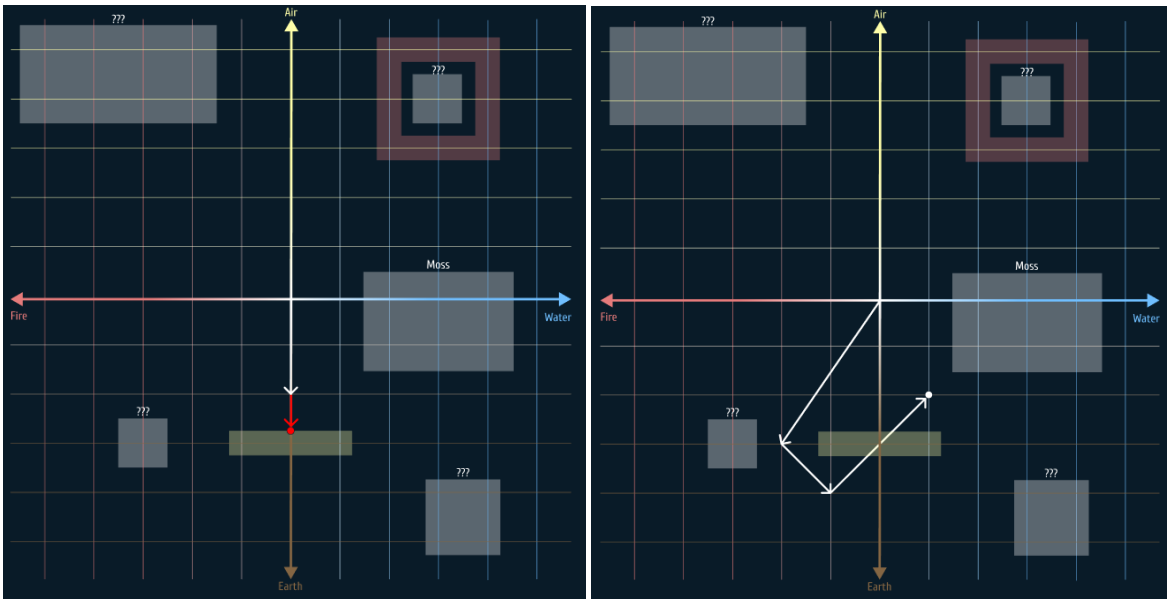
Notice that this is not the only solution: going twice right and twice down also is a valid path, albeit a longer one (4 vectors). See two examples of valid, longer paths below.





Elemental zones - First obstacles

Besides discovery zones, there are also colorful rectangles in 2D space. These are Elemental zones and can only be traversed in one direction – in the direction of their respective elements. For yellow zones, this means that the air component must be stronger than the earth component. Compare these two paths, first invalid and the other valid, for demonstration.



The path on the left is invalid, because the second ingredient's earth component is stronger than the air component (its vector is pointing in downward direction). The path on the right is valid, since the third ingredient has more air than earth energy (its vector is pointing upwards). Bigger elemental zones create one-way corridors.

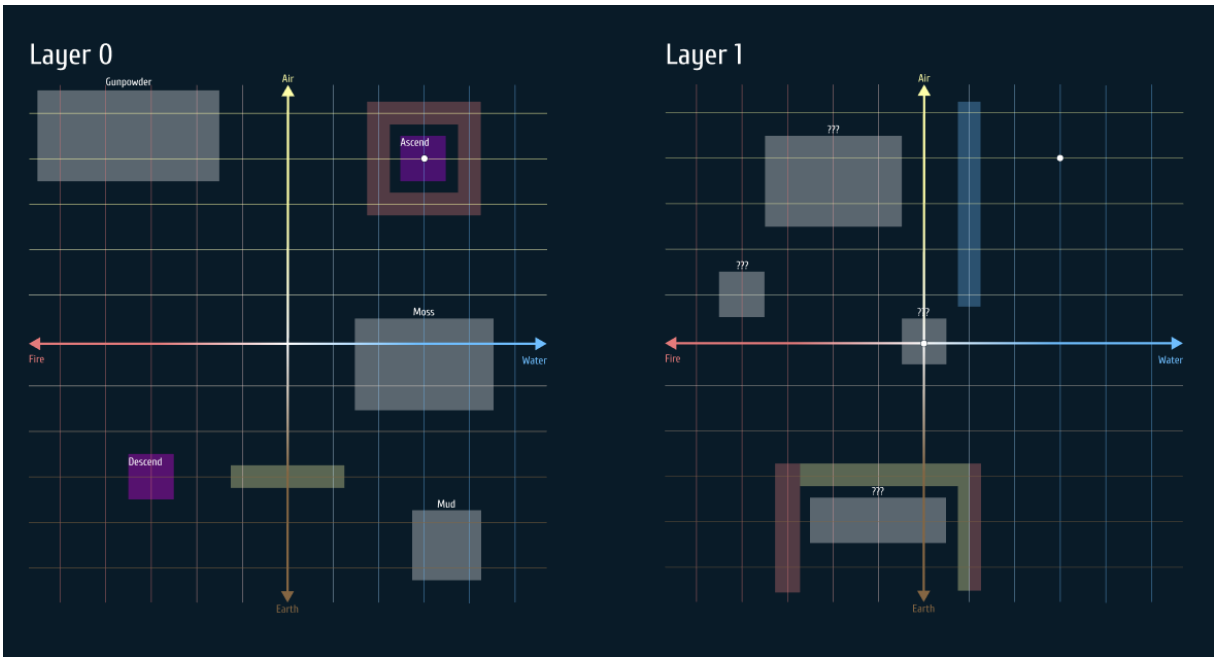
Player model

The player might reason like this:

"The game won't let me cross this zone with this ingredient. Should I just avoid those zones? The top-right red area surrounds a discovery zone, so there must be a way to get through these eventually. Let's see, this area is yellow. It might have something to do with air energy, since air is yellow. The red one should be somehow connected with fire. Is the yellow area blocking the ingredient because there is no air energy in this ingredient? Or is it pushing the vector back and I need a stronger earth ingredient? Or perhaps it's only passable in the up direction, since air is up? Or do I need to unlock another alchemy station that allows me to pass these areas?"

Aether zones – adding the Third Dimension

The fifth classical element is Aether. Ingredients cannot have Aether energy, instead the map contains special Aether zones, color-coded with purple. These zones allow the mixture to ascend and descend to another layer of the map. These should be thought of as teleporters rather than stairs, since those areas won't have a corresponding ascend/descent pair. When the mixture reaches Aether zone, it is moved to another layer and starts at the same XY coordinates. Each new layer comes with new discoveries and challenges.



Fuel - Expanding the problem space

By discovering stronger ingredients, the player can explore the alchemical map further away from the origin. However, they soon realize that using limited number of ingredients makes it very hard to navigate, with Discovery zones out of reach and blocked by Elemental zones.

The fuel mechanic aims to solve this problem. Discovering higher-grade fuel and using it on a cauldron (one-time upgrade) allows the player to use more ingredients in the mixture, thus making the paths longer and giving more freedom to the player.

Note that each extra ingredient increases the problem space exponentially, so careful balancing is required regarding when better fuel becomes available.

Ingredient processing

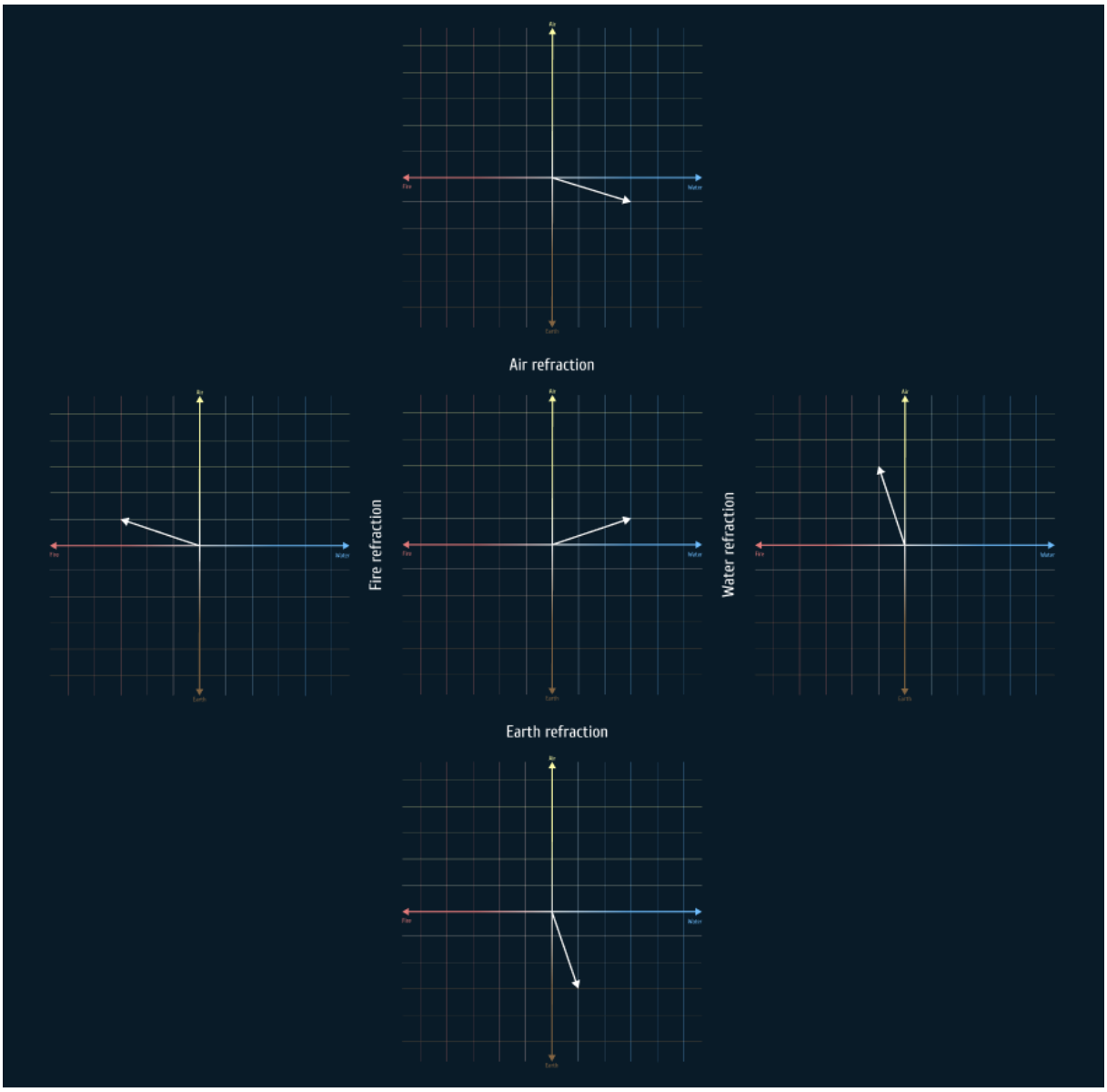
Besides the cauldron, other alchemical stations are also available. These can be used to alter the energy of ingredients in various ways. Possible transformations include:

- Pressing the ingredient to increase potency (scaling a vector by a set factor)
- Refracting ingredient's energy in a magic crystal (changing a vector's direction)
- Tempering ingredient (removing one elemental component)
- Distilling (removing all but one elemental component)

Each processing station has 4 modes of working, associated with elemental energy.

For instance, pressing the ingredient using fire mode scales its vector by a factor of 3, using air scales by a factor of 2, water by a factor of 0.5, and earth by a factor of 0.25.

Refracting an ingredient in fire mode flips its vector horizontally, in air mode - vertically, in water mode rotates the vector 90 degrees counterclockwise, in earth mode rotates 90 degrees clockwise.



The starting vector is in the middle - (3, 1). Refracting with fire (left) transforms it into (-3,1), with air (up) into (3, -1), with water (right) into (-1, 3), with earth (down) into (1, -3).

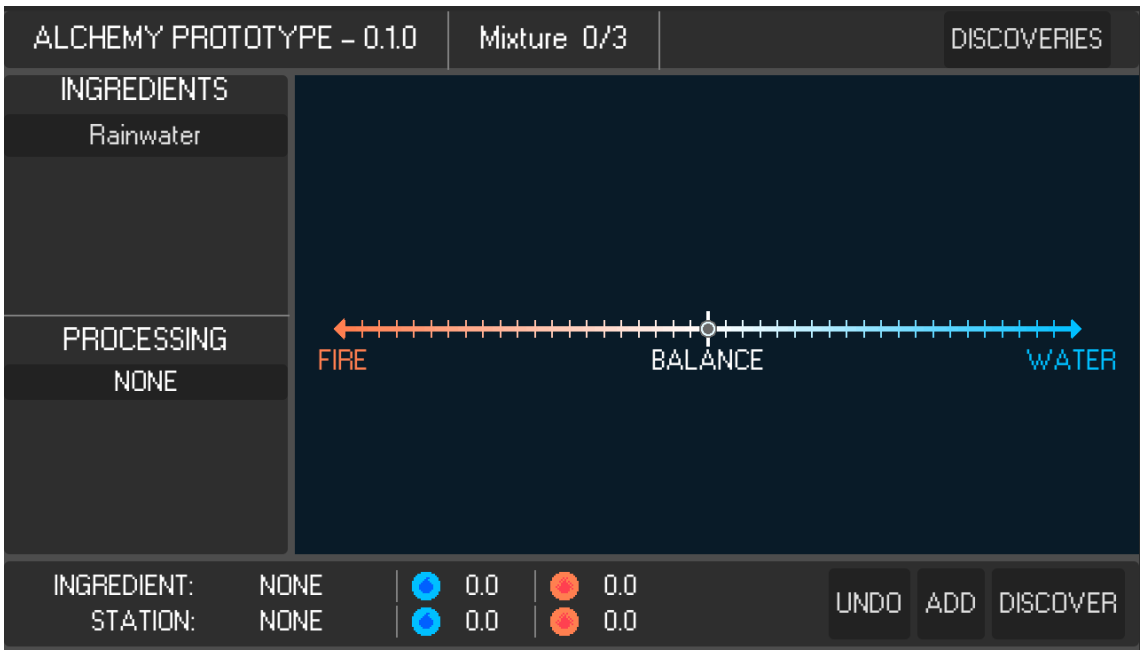
The transformations should be limited to one per ingredient to avoid exponential complexity growth.

Prototype

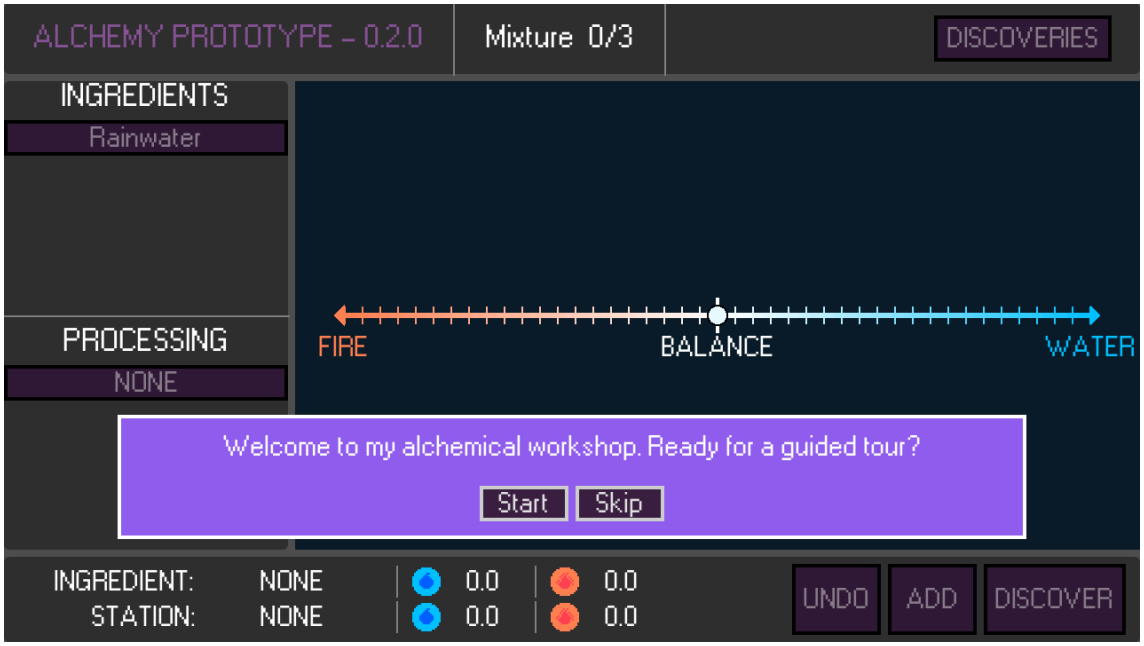
The full design eventually expands into multiple dimensions and additional processing mechanics. Before developing those systems further, I built a minimal prototype in Godot

(Github) to test whether the fundamental 1D mental model could be understood through experimentation.

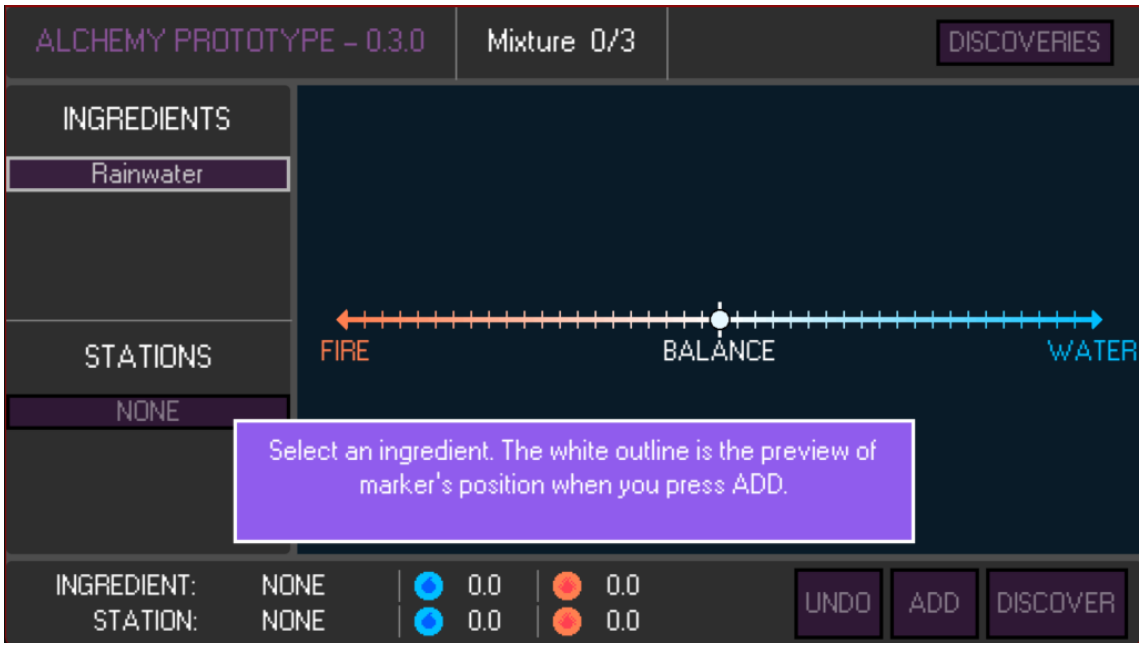
V 0.1.0



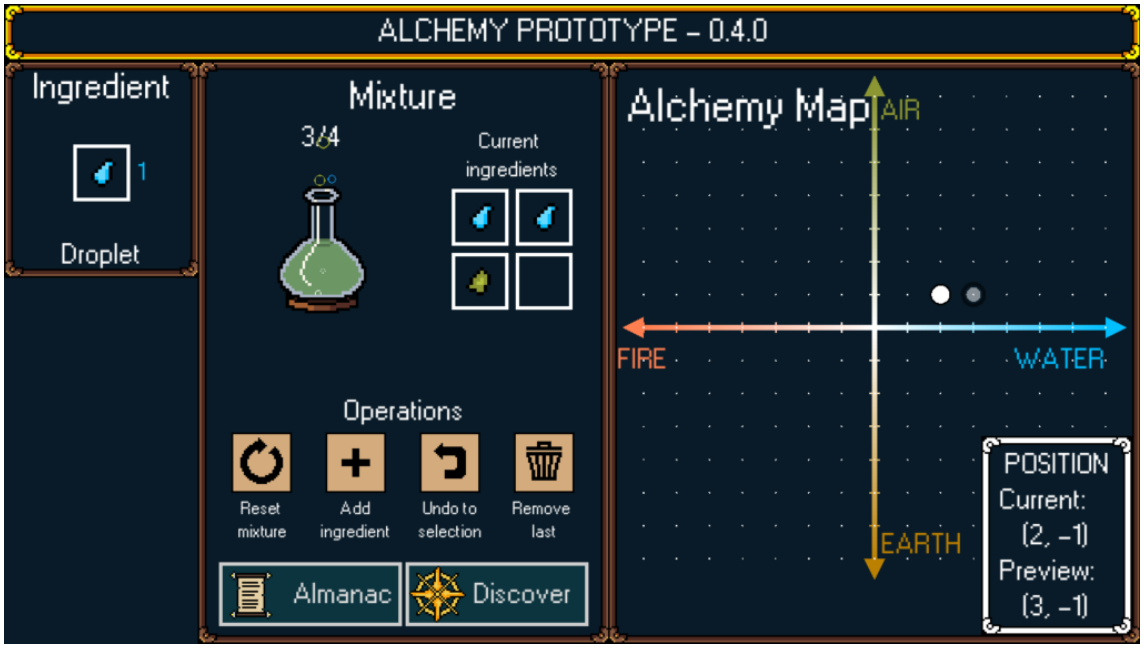
V 0.2.0



V 0.3.0



v 0.4.0



Playtesting method

I have created content for about 30 minutes of playtime and asked my friends to play the prototype. I chose three testers, their profiles:

- Hardcore gamer in 40s, male
- Casual gamer in 30s, male
- Non-gamer in 30s, female

Participants received no verbal explanation of the mechanics. They interacted with the prototype's in-game tutorial and were observed when playing. When they reached the end of content, I asked them a few questions:

1. What did you do in the game?
2. How does the game work?
3. What could be explained better?
4. Why pure water moves the marker further than rainwater?
5. Why doesn't steam move the marker at all?
6. What would be the displacement for an ingredient with 9 fire energy and 7 water energy?
7. If there were 2 additional elements – air for up, earth for down, what would the number line become?
8. Can you think of a few new ingredients and their energy in 4-elemental model? How would it correspond to marker move?

Preliminary Findings

The minimal tutorial explaining UI elements proved to be enough for players to understand the basics of alchemy system (ingredients, alchemy line, mixture, discovery). The players quickly understood the analogy to number line; two playtesters even referred to water as "positive" and fire as "negative", the other one chose "forward" and "backward". All three playtesters correctly calculated the displacement of an unfamiliar ingredient. This suggests that they understood the relationship between elemental composition and movement along the alchemy line.

The players quickly figured out they are supposed to explore the alchemy line to find new discoveries. One player commented on how the mixture capacity and maximum energy of the ingredients constrict the movement to a certain subspace. Another mentioned opening and closing the journal is tedious and they would like an option to "pin" a discovery to the main screen. All players were confused about purple discovery zones showing up to indicate this area has already been explored. One said they should show a corresponding discovery on hover.

Ingredient processing was deliberately not explained in the tutorial but hinted in the discovery journal. Unfortunately, this caused a bit of confusion regarding its use. One player assumed they have not pressed the button enough times, which indicates not enough visual feedback.

When asked to imagine two additional elemental axes, all three independently proposed a 2D coordinate system. This provides preliminary evidence that the spatial model can be extended beyond its initial 1D representation.

For a 4-elemental system, the players proposed ingredients such as:

- Pearl – 2 Air, 3 Earth
- Gold – 7 Earth, 1 Fire
- Wind – 4 Air
- Balance Stone – 2 Fire, 2 Air, 2 Water, 2 Earth

And correctly described the corresponding movement.

The playtesters said they enjoyed the prototype and commented on its short length.

Although diverse, the sample was small and consisted of friends, so these results should be treated as exploratory rather than conclusive. Further testing is needed to establish how consistently new players can discover and apply the rules without assistance.

Iteration

I have gathered observations and proposed changes in design:

Observation	Design response
Ingredient processing without any explanation is confusing the players.	Tutorial now includes basics of processing ingredients. In Inventory menu, “PROCESSING” label changed to “STATIONS”.
Discovery zones are not understood	Tutorial on discovery zones now explicitly mentions they need to be unlocked in Discovery Journal.
Not enough feedback on ingredient and processing selection	Adding an ingredient to the mixture no longer resets selected ingredient and station. Pressing on UNDO now sets the selection to undone action.
Journal access interrupted experimentation	Identified a need for more convenient access to discovery hints.

Those changes were implemented in prototype v0.3.0.

Design Questions and Validation

The core system establishes a foundation for knowledge-driven alchemy. The prototype verified some questions - primarily about the player’s model. The following questions represent key areas for further iteration and validation:

1. Learning and discovery:
    a. How much information should be provided through explicit tutorials vs learned through player experimentation?

    b. At what point should navigating the elemental space feel like satisfying problem-solving rather than hypothesizing and discovering new rules?
2. Challenge and complexity- How should ingredient capacity, discovery, elemental and Aether zones evolve to maintain meaningful problem solving without overwhelming the player?
3. Solution diversity:
    a. As players unlock more ingredients and processing stations, how can we ensure that challenges continue to support multiple viable approaches?
    b. How should discovery zones be designed to reward experimentation without encouraging brute-force ingredient combinations?
4. Failure and feedback:
    a. What feedback can help players understand why a path failed and make a more informed next attempt, without simply revealing the answer?
    b. How can we inform the player that some discovery zones are unreachable without unlocks from other systems?
5. Spatial readability:
    a. How can we represent 3D elemental space in a way that is easy to understand and navigate?
    b. What visual and interaction conventions will help players reason about elemental zones and Aether transitions?
6. Additional features - Should we allow the player to start new experiments on any previously visited layer, as opposed to the origin in zero layer?

These questions define the next stage of development - tuning the complexity curve, finalizing decision on available mechanics and ensuring that every discovery feels earned.